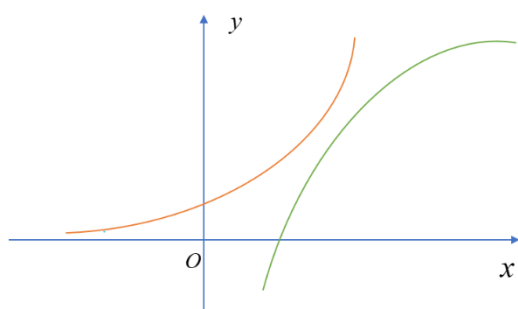


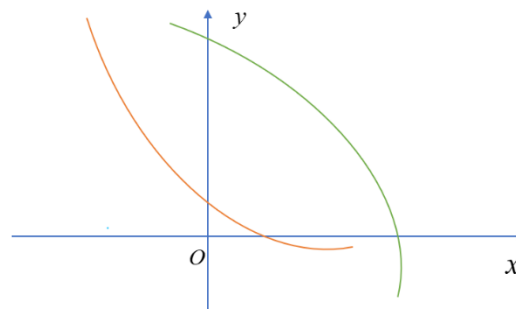
4.3 Monotonic Functions and the First Derivative Test

In sketching the graph of a differentiable function, it is useful to know where it increases and where it decreases over an interval. This section gives a test to determine where it increases and where it decreases.

① Increasing Functions and Decreasing Functions



increasing function



decreasing function

Definition

- (a) $f(x)$ is increasing on $I \Leftrightarrow f(x_1) < f(x_2)$ whenever $x_1 < x_2$.
- (b) $f(x)$ is decreasing on $I \Leftrightarrow f(x_1) > f(x_2)$ whenever $x_1 < x_2$.

- A function that is increasing or decreasing on an interval is said to be **monotonic** on the interval.

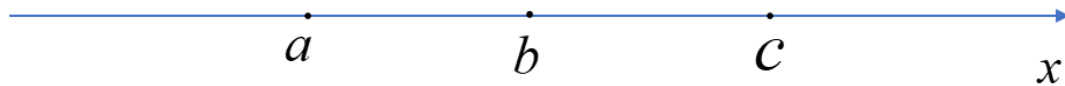
② Increasing and Decreasing Test

As another corollary to the Mean Value Theorem, we show that functions with positive derivatives are increasing functions and functions with negative derivatives are decreasing functions.

Corollary Suppose that f is continuous on $[a, b]$ and differentiable on (a, b) .

- If $f'(x) > 0$ at each point $x \in (a, b)$, then f is increasing on $[a, b]$.
- If $f'(x) < 0$ at each point $x \in (a, b)$, then f is decreasing on $[a, b]$.

To find the intervals where a function f is increasing or decreasing, we first find all of the critical points of f , such as a, b, c . Then, we use these points to divide x axis into several intervals.



The final step is to determine the sign of f' in each interval. One way we can determine the sign of f' on (a, b) is simply by evaluating the derivative at a single point c in (a, b) . If $f'(c) > 0$, then $f'(x) > 0$ for all x in (a, b) so f is increasing on $[a, b]$; if $f'(c) < 0$, then f is decreasing on $[a, b]$. It doesn't matter which point c we choose in (a, b) , since the sign of $f'(c)$ is the same for all choices.

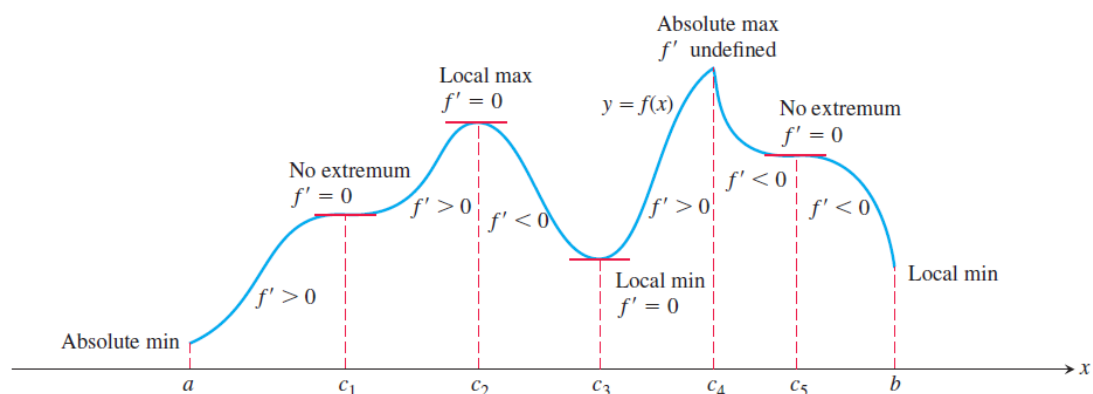
Example 1 Find the critical points of $f(x) = x^3 - 12x - 5$ and identify the open intervals on which f is increasing and on which f is decreasing.

Example 2 Determine where the function

$$f(x) = x^3 - \frac{3}{2}x^2 - 6x + 3$$

is increasing and where it is decreasing.

③ First Derivative Test for Local Extrema

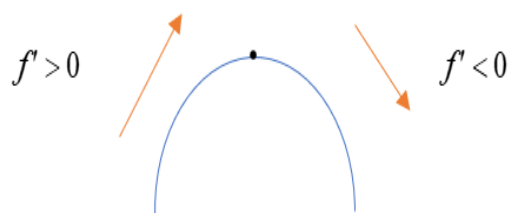


In the above Figure, at the points where f has a minimum value, $f' < 0$ immediately to the left and $f' > 0$ immediately to the right. (If the point is an endpoint, there is only one side to consider.) Thus, the function is decreasing on the left of the minimum value and it is increasing on its right. Similarly, at the points where f has a maximum value, $f' > 0$ immediately to the left and $f' < 0$ immediately to the right. Thus, the function is increasing on the left of the maximum value and decreasing on its right. In summary, at a local extreme point, the sign of $f'(x)$ changes.

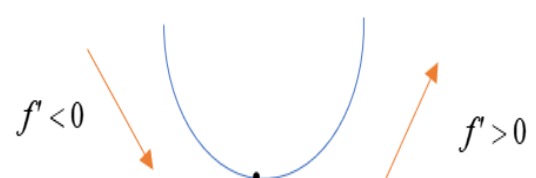
● First Derivative Test for Local Extrema

Suppose that c is a critical point of a continuous function f , and that f is differentiable at every point in some interval containing c except possibly at c itself. Moving across this interval from left to right,

1. if f' changes from negative to positive at c , then f has a local minimum at c ;
2. if f' changes from positive to negative at c , then f has a local maximum at c ;
3. if f' does not change sign at c , then f has no local extremum at c .



local max

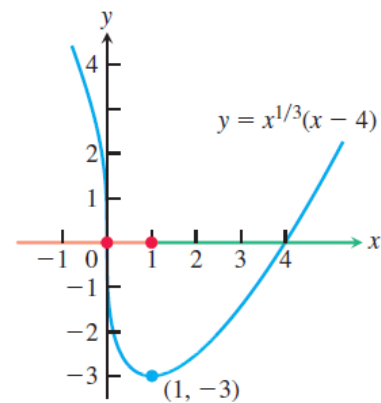


local min

Example 3 Find the critical points of

$$f(x) = x^{\frac{1}{3}}(x - 4) = x^{\frac{4}{3}} - 4x^{\frac{1}{3}}.$$

Identify the open intervals on which f is increasing and decreasing. Find the function's local and absolute extreme values.



Example 4 Within the interval $[0, 2\pi]$ find the critical points of

$$f(x) = \sin^2 x - \sin x - 1.$$

Identify the open intervals on which f is increasing and decreasing. Find the function's local and absolute extreme values.

